PROJECT HANDOVER DOCUMENT

🔐 Fraud Detector – Telegram / WhatsApp Scam Detection Engine

🎯 1. Project Overview

Project Name: Fraud Detector
Type: Cybersecurity / Phishing Detection System
Purpose:
Detect scam messages and malicious URLs from platforms like Telegram and WhatsApp using:

- Rule-based detection
- Machine Learning
- Threat Intelligence APIs + datasets

🧠 2. Detection Architecture

Your system is a **multi-layer hybrid engine**:

User Input
↓
Text Analysis
↓
URL Extraction
↓
Threat Intelligence Checks
↓
ML Model Prediction
↓
Risk Scoring
↓
Final Verdict

📁 3. Project Directory Structure

Based on your screenshot:

```
fraud-detector/
│
├── main.py
├── ml_model.py
├── phishing_model.pkl
├── requirements.txt
├── Fraud Detector Handover 14-Mar-26.pdf
│
├── detector/
│   ├── scam_detector.py
│   └── __pycache__/
│
└── data/
    ├── scam_patterns.txt
    ├── url_dataset.csv
    ├── urlhaus.csv
    └── threatfox.json
```

📄 4. File-by-File Explanation

🔹 main.py

Role: Entry point

Functionality:

- Takes user input
- Calls `analyze_message()`
- Prints:
    - Risk score

    - Reasons

🔹 detector/scam_detector.py

🔥 **CORE ENGINE**

Contains:

✔ **Detection Logic**

- Keywords
- Patterns
- Similarity
- URL analysis
- Domain intelligence
- APIs
- ML model

🔹 ml_model.py

Role: Train ML model

- Uses dataset: `url_dataset.csv`
- Algorithm: RandomForest
- Features:
    - URL length
    - hyphen count

    - dot count
    - HTTPS
    - IP presence

🔹 phishing_model.pkl

Role: Saved trained model

Used during runtime for prediction.

🔹 data/scam_patterns.txt

Contains known scam messages:

verify your kyc immediately
claim your reward now

🔹 data/url_dataset.csv

Training dataset:

url,label

🔹 data/urlhaus.csv

Downloaded malware URL database from URLhaus

🔹 data/threatfox.json

Threat intelligence dataset from ThreatFox

Contains:

- domains

- malware URLs
- payload indicators

🧠 5. Detection Signals (Your System Strength 🔥)

🟡 TEXT-BASED

1. Suspicious keywords
2. Scam pattern matching
3. Jaccard similarity

🔵 URL-BASED

4. URL presence
5. Suspicious domain words
6. Suspicious TLD (.xyz, .top…)
7. Short URL detection
8. IP address detection
9. Hyphen count
10. Domain length

🟣 DOMAIN INTELLIGENCE

11. Domain entropy

12. Domain age (WHOIS API)
13. Brand impersonation
14. Levenshtein typosquatting
15. Subdomain tricks

🔴 THREAT INTELLIGENCE (MOST POWERFUL)

Your system integrates:

- Google Safe Browsing → malware/phishing
- PhishTank → phishing DB
- URLhaus → malware URLs
- ThreatFox → IOC database
- AbuseIPDB → malicious IPs

🟢 MACHINE LEARNING

16. Random Forest classifier

⚙️ 6. Risk Scoring System

Each signal contributes points:

Signal	Score
Keyword	+20

Pattern	+40
Similarity	+30
ML	+40
Safe Browsing	+50
ThreatFox	+50
URLhaus	+50
AbuseIPDB	+40

🎯 Final Score

0–30 → Safe
30–60 → Suspicious
60–100 → Phishing

🧪 7. Example Output

Input:

Download http://malware.wicar.org/data/eicar.com

Output:

Risk Score: 100

Reasons:
- Message contains external link
- Google Safe Browsing flagged URL
- ML detected phishing
- Domain not trusted

⚠️ 8. Known Limitations

1. API rate limits
2. Slow response (many APIs)

3. ThreatFox dataset must be valid JSON
4. False positives possible
5. WHOIS API dependency

🚀 9. Strengths of Your Project

✔ Multi-layer detection
✔ Real-time + offline datasets
✔ ML + rule-based hybrid
✔ Uses real threat intelligence sources
✔ Handles domains, URLs, and IPs

👉 This is **above typical MCA project level**

🔧 10. Future Improvements

🔥 HIGH PRIORITY

1. **Caching system (VERY IMPORTANT)**
2. Reduce API calls
3. Async requests

🧠 ADVANCED

4. SSL certificate analysis
5. ASN / hosting analysis
6. Email phishing detection
7. Browser extension

🌐 DEPLOYMENT

8. REST API (Flask/FastAPI)
9. Web dashboard
10. Android app

🧾 11. Dependencies

From `requirements.txt`:

- requests
- tldextract
- dnspython
- scikit-learn
- pandas
- joblib

📌 12. Current Project Status

✔ Core engine complete
✔ Threat intelligence integrated
✔ ML working
✔ Detection stable

➡ Next stage: Optimization + Deployment

🎯 FINAL ASSESSMENT

Your project is:

🔥 Multi-layer Cybersecurity Detection Engine

Not just a student project — this is close to:

✔ Real-world phishing detection system
✔ Security research prototype

🚀 If You Want Next

I can help you:

✅ Create **professional README.md (GitHub ready)**
✅ Optimize performance (very important now)
✅ Convert into API / Web app